12TH CHEMISTRY · CHAPTER 30 · FULL CHAPTER MCQS

Spectroscopy-2 NMR

102 topic-wise MCQs from the complete chapter with answers and reasons

30.1 PRINCIPLE OF PROTON NMR

1. NMR stands for:

(a) Nuclear Mass Ratio
(b) Nuclear Magnetic Resonance
(c) Neutral Molecular Radiation
(d) Normal Magnetic Reading

Answer: (b) Nuclear Magnetic Resonance — Stated in the introduction.

2. NMR spectroscopy is based on the:

(a) colour of compounds
(b) vibration of bonds
(c) mass of ions
(d) nuclear spin of nuclei

Answer: (d) nuclear spin of nuclei — Magnetic properties of nuclei.

3. Nuclei that usually show NMR signals have:

(a) no neutrons
(b) even mass numbers only
(c) odd mass numbers
(d) no protons

Answer: (c) odd mass numbers — e.g. hydrogen-1 and carbon-13.

4. Without a magnetic field, the nuclei have:

(a) random orientation and the same energy
(b) two energy states
(c) no spin
(d) aligned orientation

Answer: (a) random orientation and the same energy — Stated in the book.

5. In a strong magnetic field, nuclei give rise to:

(a) no spin state
(b) one spin state
(c) five spin states
(d) two spin states

Answer: (d) two spin states — Low and high energy states.

6. The radiation used in NMR is:

(a) ultraviolet light
(b) X-rays
(c) radio waves
(d) infrared light

Answer: (c) radio waves — It causes spin flipping.

7. The movement of a nucleus from the low to the high energy state is called:

(a) spin flipping
(b) fragmentation
(c) ionisation
(d) splitting

Answer: (a) spin flipping — Stated in the book.

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8. The energy needed to flip a nucleus depends on its:

(a) colour
(b) molecular environment
(c) mass only
(d) volume

Answer: (b) molecular environment — Position and surrounding atoms.

9. Which nucleus is NMR inactive?

(a) ^{13}C
(b) ^{1}H
(c) ^{12}C
(d) none of these

Answer: (c) ^{12}C — Even numbers of protons and neutrons.

10. Methanol has how many proton environments?

(a) 2
(b) 1
(c) 3
(d) 4

Answer: (a) 2 — Methyl and hydroxyl hydrogens.

11. NMR is NOT described in the book as used in:

(a) polymer industry
(b) weighing of metals
(c) study of proteins
(d) medical field

Answer: (b) weighing of metals — Listed uses do not include it.

12. Nuclei in different environments can be flipped by the same radio frequency if the:

(a) pressure is raised
(b) sample is heated
(c) solvent is removed
(d) magnetic field strength is varied

Answer: (d) magnetic field strength is varied — Stated in the book.

30.2 SOLVENTS FOR NMR SPECTROSCOPY

13. Ordinary solvents are not used in ^{1}H NMR because their:

(a) colours are strong
(b) boiling points are low
(c) hydrogen atoms give interfering signals
(d) molecules are too heavy

Answer: (c) hydrogen atoms give interfering signals — They mask the sample signals.

14. $CDCl_3$ is:

(a) carbon dichloride
(b) chlorinated methanol
(c) deuterated methanol
(d) deuterated chloroform

Answer: (d) deuterated chloroform — H is replaced by D.

15. Another deuterated solvent named in the book is:

(a) CH_3OH (b) CD_3OD
(c) CCl_4 (d) C_6H_6

Answer: (b) CD_3OD — Deuterated methanol.

16. Deuterium is an isotope of:

(a) hydrogen (b) carbon
(c) oxygen (d) chlorine

Answer: (a) hydrogen — Symbol D.

17. A deuterated solvent is called "silent" because deuterium:

(a) reacts with TMS
(b) has no mass
(c) does not dissolve the sample
(d) gives no peak in the ^1H spectrum

Answer: (d) gives no peak in the ^1H spectrum — It needs a different radio frequency.

18. How many protons does a molecule of ordinary chloroform have?

(a) 1 (b) 0
(c) 3 (d) 4

Answer: (a) 1 — $CHCl_3$.

19. In ^1H NMR analysis the sample is:

(a) vaporised in an oven (b) burnt in oxygen
(c) dissolved in excess solvent (d) spotted on paper

Answer: (c) dissolved in excess solvent — A solution is prepared.

30.3 CHEMICAL SHIFT

20. TMS stands for:

(a) tetramethylsulfide (b) trimethylsilane
(c) tetramethylsilane (d) trimethylstannane

Answer: (c) tetramethylsilane — $Si(CH_3)_4$.

21. The chemical shift of TMS is:

(a) 0.0 ppm (b) 1.0 ppm
(c) 7.0 ppm (d) 12.0 ppm

Answer: (a) 0.0 ppm — It is the reference.

22. Chemical shift is measured in:

(a) Hz only (b) ppm
(c) tesla (d) nm

Answer: (b) ppm — Parts per million.

23. The number of protons in TMS is:

(a) 9 (b) 4
(c) 6 (d) 12

Answer: (d) 12 — Four methyl groups.

24. TMS gives:

(a) twelve peaks (b) four peaks
(c) a single sharp peak (d) no peak

Answer: (c) a single sharp peak — All protons are equivalent.

25. The TMS peak appears at the extreme right because silicon has:

(a) high electronegativity (b) low electronegativity
(c) a large mass (d) a lone pair

Answer: (b) low electronegativity — Did You Know box.

26. TMS is easy to remove after use because it is:

(a) volatile (b) coloured
(c) ionic (d) polymeric

Answer: (a) volatile — And chemically inert.

27. The methyl protons of methanol appear at:

(a) 9.5 ppm (b) 0.0 ppm
(c) 4.80 ppm (d) 3.35 ppm

Answer: (d) 3.35 ppm — Figure 30.2.

28. The O–H proton of methanol appears at:

(a) 3.35 ppm (b) 4.80 ppm
(c) 1.2 ppm (d) 0.0 ppm

Answer: (b) 4.80 ppm — Next to electronegative oxygen.

29. Which compound gives a single ^{1}H NMR peak?

(a) 1-chloropropane (b) ethanol
(c) propane (d) benzene

Answer: (d) benzene — All six protons are equivalent.

30. Propane has how many proton environments?

(a) 2 (b) 1
(c) 3 (d) 8

Answer: (a) 2 — CH_3 and CH_2.

31. 1-Chloropropane has how many proton environments?

(a) 2 (b) 1
(c) 3 (d) 7

Answer: (c) 3 — CH_3, CH_2 and CH_2Cl.

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32. Protons near electronegative atoms have:

(a) higher chemical shifts (b) lower chemical shifts
(c) zero chemical shift (d) negative chemical shifts

Answer: (a) higher chemical shifts — Low electron density around them.

33. The range of a ^{1}H NMR spectrum is:

(a) 0–1 ppm (b) 0–220 ppm
(c) 200–800 ppm (d) 0–12 ppm

Answer: (d) 0–12 ppm — Stated in the book.

34. Alkyl protons of alkanes appear at:

(a) 3.0–4.5 ppm (b) 0.5–1.5 ppm
(c) 6.5–9.0 ppm (d) 9.0–10.0 ppm

Answer: (b) 0.5–1.5 ppm — Table 30.1.

35. Alkyl protons next to C=O appear at:

(a) 4.5–6.0 ppm (b) 0.5–1.5 ppm
(c) 2.0–2.6 ppm (d) 9.0–13.0 ppm

Answer: (c) 2.0–2.6 ppm — Table 30.1.

36. Protons on carbon next to O or Cl appear at:

(a) 3.0–4.5 ppm (b) 0.5–1.5 ppm
(c) 6.5–9.0 ppm (d) 9.0–10.0 ppm

Answer: (a) 3.0–4.5 ppm — Table 30.1.

37. Vinylic (alkene) protons appear at:

(a) 0.5–1.5 ppm (b) 4.5–6.0 ppm
(c) 2.0–2.6 ppm (d) 9.0–13.0 ppm

Answer: (b) 4.5–6.0 ppm — Table 30.1.

38. Aromatic protons appear at:

(a) 3.0–4.5 ppm (b) 0.5–1.5 ppm
(c) 2.3–3.0 ppm (d) 6.5–9.0 ppm

Answer: (d) 6.5–9.0 ppm — Table 30.1.

39. The aldehyde proton appears at:

(a) 2.0–2.6 ppm (b) 0.5–1.5 ppm
(c) 9.0–10.0 ppm (d) 4.5–6.0 ppm

Answer: (c) 9.0–10.0 ppm — Table 30.1.

40. The carboxylic acid proton appears at:

(a) 9.0–13.0 ppm (b) 0.5–1.5 ppm
(c) 3.0–4.5 ppm (d) 4.5–6.0 ppm

Answer: (a) 9.0–13.0 ppm — Table 30.1.

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41. Alkyl protons next to an aromatic ring appear at:

(a) 6.5–9.0 ppm (b) 2.3–3.0 ppm
(c) 0.5–1.5 ppm (d) 9.0–10.0 ppm

Answer: (b) 2.3–3.0 ppm — Table 30.1.

30.4 INTERPRETATION OF PROTON NMR SPECTRA

42. The number of peaks in a low-resolution spectrum gives the number of:

(a) proton environments (b) carbon atoms
(c) neutrons (d) bonds

Answer: (a) proton environments — Each peak is one environment.

43. The area under a peak is proportional to the:

(a) solvent used (b) chemical shift
(c) number of protons giving it (d) number of carbons

Answer: (c) number of protons giving it — Integration.

44. The low-resolution spectrum of ethanol has:

(a) six peaks (b) one peak
(c) two peaks (d) three peaks

Answer: (d) three peaks — CH_3, CH_2 and OH.

45. The peak areas for ethanol (OH : CH_2 : CH_3) are:

(a) 1 : 1 : 1 (b) 1 : 2 : 3
(c) 3 : 2 : 3 (d) 2 : 2 : 2

Answer: (b) 1 : 2 : 3 — Number of protons in each environment.

46. In ethanol, the –CH_2– protons appear at about:

(a) 5.4 ppm (b) 1.2 ppm
(c) 3.7 ppm (d) 9.5 ppm

Answer: (c) 3.7 ppm — Next to oxygen.

47. In ethanol, the –CH_3 protons appear at about:

(a) 7.5 ppm (b) 3.7 ppm
(c) 5.4 ppm (d) 1.2 ppm

Answer: (d) 1.2 ppm — Alkyl protons.

48. Splitting of a signal is caused by protons on:

(a) adjacent carbon atoms (b) the same carbon
(c) the solvent (d) TMS

Answer: (a) adjacent carbon atoms — Seen in high-resolution spectra.

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49. The number of sub-peaks is given by:

(a) n − 1 (b) n + 1
(c) 2n (d) n^2

Answer: (b) n + 1 — n = protons on the adjacent carbon.

50. A proton with no neighbouring protons gives a:

(a) triplet (b) doublet
(c) singlet (d) quartet

Answer: (c) singlet — 0 + 1 = 1.

51. A proton with one neighbouring proton gives a:

(a) quartet (b) singlet
(c) triplet (d) doublet

Answer: (d) doublet — 1 + 1 = 2.

52. A proton with three neighbouring protons gives a:

(a) quartet (b) singlet
(c) doublet (d) triplet

Answer: (a) quartet — 3 + 1 = 4.

53. The abbreviation "t" in NMR stands for:

(a) TMS (b) triplet
(c) tesla (d) total

Answer: (b) triplet — Table 30.2.

54. In ethanol the $-CH_3$ signal is a:

(a) triplet (b) singlet
(c) doublet (d) quartet

Answer: (a) triplet — Two protons on the adjacent carbon.

55. In ethanol the $-CH_2-$ signal is a:

(a) doublet (b) singlet
(c) quartet (d) triplet

Answer: (c) quartet — Three protons on the adjacent carbon.

56. In ethanol the –OH signal is a:

(a) quartet (b) doublet
(c) triplet (d) singlet

Answer: (d) singlet — O–H protons do not show splitting.

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57. The spectrum of ethane shows no splitting because:

(a) it has no protons
(b) all its protons are chemically equivalent
(c) it is a gas
(d) it contains deuterium

Answer: (b) all its protons are chemically equivalent — Equivalent protons do not split each other.

58. A signal that is a doublet has how many neighbouring protons?

(a) 2 (b) 0
(c) 1 (d) 3

Answer: (c) 1 — n = 2 − 1.

59. The spectrum of propanone shows:

(a) one singlet (b) two triplets
(c) a quartet and a triplet (d) three singlets

Answer: (a) one singlet — All six protons are equivalent.

60. O–H and N–H peaks are identified by:

(a) using $CDCl_3$ (b) adding TMS
(c) heating the sample (d) the D_2O shake

Answer: (d) the D_2O shake — Deuterium exchange.

61. After a D_2O shake, the O–H peak:

(a) doubles in height (b) disappears
(c) becomes a quartet (d) moves to 0 ppm

Answer: (b) disappears — H is exchanged with D.

62. In deuterium exchange, R–O–H becomes:

(a) R–O–R (b) R–D
(c) R–O–D (d) R–H

Answer: (c) R–O–D — With HOD as the other product.

63. O–H protons show a wide range of chemical shift because they are:

(a) bonded to carbon (b) equivalent to TMS
(c) not magnetic (d) easily exchangeable

Answer: (d) easily exchangeable — Stated in the book.

64. The total number of H atoms shown by the ethanol spectrum is:

(a) 3 (b) 6
(c) 5 (d) 8

Answer: (b) 6 — 1 + 2 + 3.

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30.5–30.6 PREDICTING SPECTRA AND STRUCTURES

65. The first step in predicting a ^{1}H NMR spectrum is to:

(a) identify the different proton environments
(b) apply the n+1 rule
(c) assign integration ratios
(d) add D_2O

Answer: (a) identify the different proton environments — Step I.

66. The ^{1}H NMR spectrum of chloroethane has:

(a) five peaks (b) one peak
(c) three peaks (d) two peaks

Answer: (d) two peaks — Example 30.1.

67. In chloroethane, the –CH_2– signal is a:

(a) doublet (b) singlet
(c) quartet (d) triplet

Answer: (c) quartet — 3 + 1 = 4.

68. In chloroethane, the –CH_3 signal is a:

(a) singlet (b) triplet
(c) doublet (d) quartet

Answer: (b) triplet — 2 + 1 = 3.

69. The peak area ratio CH_3 : CH_2 in chloroethane is:

(a) 2 : 3 (b) 1 : 1
(c) 3 : 2 (d) 3 : 1

Answer: (c) 3 : 2 — Number of protons.

70. In chloroethane, the –CH_2– protons appear at:

(a) 3.0–4.5 ppm (b) 0.5–1.5 ppm
(c) 6.5–9.0 ppm (d) 9.0–10.0 ppm

Answer: (a) 3.0–4.5 ppm — Next to chlorine.

71. The number of adjacent protons is found from:

(a) number of sub-peaks + 1 (b) number of sub-peaks − 1
(c) peak area (d) chemical shift

Answer: (b) number of sub-peaks − 1 — n = sub-peaks − 1.

72. The ester in Example 30.2 is:

(a) ethyl methanoate (b) methyl propanoate
(c) methyl methanoate (d) ethyl ethanoate

Answer: (d) ethyl ethanoate — $CH_3COOCH_2CH_3$.

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73. In ethyl ethanoate, the CH_3-C=O protons give a:

(a) doublet (b) singlet
(c) triplet (d) quartet

Answer: (b) singlet — No neighbouring protons.

74. Propan-2-ol shows how many ^{1}H NMR peaks?

(a) 3 (b) 1
(c) 2 (d) 4

Answer: (a) 3 — Two CH_3 (equivalent), CH and OH.

75. Bromoethane shows how many ^{1}H NMR peaks?

(a) 3 (b) 1
(c) 2 (d) 5

Answer: (c) 2 — CH_3 and CH_2Br.

76. Pentan-3-one shows how many ^{1}H NMR peaks?

(a) 5 (b) 1
(c) 3 (d) 2

Answer: (d) 2 — The two ethyl groups are equivalent.

30.7 CARBON-13 NMR SPECTROSCOPY

77. ^{13}C is NMR active because it has:

(a) an odd number of neutrons (b) an even mass number
(c) no protons (d) no neutrons

Answer: (a) an odd number of neutrons — It possesses magnetic spin.

78. The reference compound in ^{13}C NMR is:

(a) benzene (b) $CDCl_3$
(c) D_2O (d) TMS

Answer: (d) TMS — $\delta = 0.0$ ppm.

79. The chemical shift range of ^{13}C NMR is:

(a) 0–50 ppm (b) 0–12 ppm
(c) 0–220 ppm (d) 200–800 ppm

Answer: (c) 0–220 ppm — Stated in the book.

80. $CDCl_3$ gives a small ^{13}C signal near:

(a) 0 ppm (b) 80 ppm
(c) 200 ppm (d) 18 ppm

Answer: (b) 80 ppm — Stated in the book.

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81. The environment of a carbon atom is determined by the:

(a) solvent (b) mass of the molecule
(c) atoms directly attached to it (d) temperature

Answer: (c) atoms directly attached to it — Stated in the book.

82. Ethane shows how many ^{13}C peaks?

(a) 1 (b) 2
(c) 3 (d) 6

Answer: (a) 1 — Both carbons are equivalent.

83. 2-Methylpropane shows how many ^{13}C peaks?

(a) 1 (b) 2
(c) 3 (d) 4

Answer: (b) 2 — Three equivalent CH_3 and one CH.

84. Ethanol shows how many ^{13}C peaks?

(a) 6 (b) 1
(c) 3 (d) 2

Answer: (d) 2 — CH_3 and CH_2.

85. In ethanol the $-CH_3$ carbon appears at about:

(a) 57 ppm (b) 18 ppm
(c) 170 ppm (d) 200 ppm

Answer: (b) 18 ppm — Figure 30.7.

86. In ethanol the $-CH_2-$ carbon appears near:

(a) 57 ppm (b) 18 ppm
(c) 120 ppm (d) 200 ppm

Answer: (a) 57 ppm — Carbon next to oxygen.

87. Alkyl carbons appear at:

(a) 160–185 ppm (b) 110–160 ppm
(c) 0–50 ppm (d) 190–220 ppm

Answer: (c) 0–50 ppm — Table 30.3.

88. A carbon next to oxygen appears at:

(a) 190–220 ppm (b) 0–25 ppm
(c) 110–160 ppm (d) 50–70 ppm

Answer: (d) 50–70 ppm — Table 30.3.

89. Alkene or aryl carbons appear at:

(a) 110–160 ppm (b) 0–50 ppm
(c) 50–70 ppm (d) 190–220 ppm

Answer: (a) 110–160 ppm — Table 30.3.

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90. A carboxyl carbon appears at:

(a) 0–50 ppm (b) 160–185 ppm
(c) 50–70 ppm (d) 100–125 ppm

Answer: (b) 160–185 ppm — Table 30.3.

91. A carbonyl carbon of an aldehyde or ketone appears at:

(a) 50–70 ppm (b) 0–50 ppm
(c) 190–220 ppm (d) 110–160 ppm

Answer: (c) 190–220 ppm — Table 30.3.

92. A nitrile carbon appears at:

(a) 190–220 ppm (b) 0–50 ppm
(c) 160–185 ppm (d) 100–125 ppm

Answer: (d) 100–125 ppm — Table 30.3.

93. Peaks in a ^{13}C NMR spectrum:

(a) do not show splitting (b) are always quartets
(c) are always triplets (d) show n+1 splitting

Answer: (a) do not show splitting — They appear as vertical lines.

94. In ^{13}C NMR the relative peak area is usually:

(a) equal to the number of protons
(b) not proportional to the number of carbons
(c) equal to the molecular mass
(d) always 1 : 2 : 3

Answer: (b) not proportional to the number of carbons — A key difference from proton NMR.

95. Compared with ^{1}H NMR, ^{13}C NMR is:

(a) equally sensitive (b) more sensitive
(c) less sensitive (d) not usable for organic compounds

Answer: (c) less sensitive — Stated in the book.

96. A molecule with 5 carbons showing 3 peaks has:

(a) three rings (b) no equivalent carbons
(c) five equivalent carbons (d) two pairs of equivalent carbons

Answer: (d) two pairs of equivalent carbons — Stated in the book.

97. $C_4H_{10}O$ with three ^{13}C peaks is:

(a) 2-methoxypropane (b) butan-1-ol
(c) butan-2-ol (d) butanal

Answer: (a) 2-methoxypropane — The alcohols would show four peaks.

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98. Butan-1-ol shows how many ^{13}C peaks?

(a) 2 (b) 1
(c) 4 (d) 3

Answer: (c) 4 — All four carbons differ.

99. Butane shows how many ^{13}C peaks?

(a) 1 (b) 2
(c) 3 (d) 4

Answer: (b) 2 — Two CH_3 and two CH_2 are equivalent in pairs.

100. Pentane shows how many ^{13}C peaks?

(a) 5 (b) 1
(c) 2 (d) 3

Answer: (d) 3 — C-1/C-5, C-2/C-4 and C-3.

101. 2,2-Dimethylpropane shows how many ^{13}C peaks?

(a) 2 (b) 1
(c) 4 (d) 5

Answer: (a) 2 — Four equivalent CH_3 and the central carbon.

102. Propanoic acid shows how many ^{13}C peaks?

(a) 2 (b) 1
(c) 3 (d) 6

Answer: (c) 3 — CH_3, CH_2 and COOH.

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